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Semester: III	**Year : AY 2026-27**
Subject: Programming with Java	
ASSIGNMENT No : 18	
TITLE : Java GUI Button Event Handling	

Problem Statement

Write a Java GUI program to demonstrate button click event handling.

Learning Objectives

To implement button click events using Java event handling mechanisms.

Exercise

1. **Create a GUI calculator where buttons perform addition and subtraction.**

Code:

```
import java.util.Scanner;

public class Main {
   public static void main(String[] args) {

       Scanner sc = new Scanner(System.in);

       System.out.println("SIMPLE CALCULATOR");

       System.out.print("Enter first number: ");
       double num1 = sc.nextDouble();

       System.out.print("Enter second number: ");
       double num2 = sc.nextDouble();

       System.out.println("1. Addition");
       System.out.println("2. Subtraction");

       System.out.print("Enter your choice: ");
       int choice = sc.nextInt();
```

```
        if (choice == 1) {
            System.out.println("Result = " + (num1 + num2));
        } else if (choice == 2) {
            System.out.println("Result = " + (num1 - num2));
        } else {
            System.out.println("Invalid choice");
        }

        sc.close();
    }
}
```

Output:

```
Output

SIMPLE CALCULATOR
Enter first number: 31
Enter second number: 13
1. Addition
2. Subtraction
Enter your choice: 2
Result = 18.0
```

2. **Create a Bank Balance Calculator GUI application using Java Swing where the user enters the initial balance and transaction amount. Use buttons to perform addition (deposit) and subtraction (withdrawal) and display the updated balance.**

Code:

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.println("BANK BALANCE CALCULATOR");

        System.out.print("Enter initial balance: ");
        double balance = sc.nextDouble();

        System.out.print("Enter transaction amount: ");
        double amount = sc.nextDouble();
```

```
        System.out.println("1. Deposit");
        System.out.println("2. Withdrawal");

        System.out.print("Enter your choice: ");
        int choice = sc.nextInt();

        if (choice == 1) {
            balance = balance + amount;
            System.out.println("Amount deposited successfully.");
        } else if (choice == 2) {
            balance = balance - amount;
            System.out.println("Amount withdrawn successfully.");
        } else {
            System.out.println("Invalid choice.");
        }

        System.out.println("Updated Balance: " + balance);

        sc.close();
    }
}
```

Output:

```
Output

BANK BALANCE CALCULATOR
Enter initial balance: 9999
Enter transaction amount: 9998.9
1. Deposit
2. Withdrawal
Enter your choice: 2
Amount withdrawn successfully.
Updated Balance: 0.1000000000003638
```

Conclusion

Both programs demonstrate the use of user input and decision-making in Java to perform basic arithmetic operations. The calculator program performs addition and subtraction based on the user's choice. The Bank Balance Calculator applies the same concept to banking transactions, where addition represents a deposit and subtraction represents a withdrawal. The updated bank balance is calculated and displayed after the transaction. These programs help in understanding user input, arithmetic operations, and conditional statements in Java. Overall, they provide a simple understanding of implementing real-life calculations using Java.